Disjoint multipair factor-three resonance graph and origin-Hessian gap

TECT verification-first repository
claim `A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION`

first issued 2026-08-03 · this version issued 2026-08-03 · v1.0

1. Purpose and scope: a restricted global Hessian really closes

R-151 proves a positive origin-Hessian gap for one adapted antipodal $p : 2p$ pair. R-152 correctly warns that positive pairwise gaps cannot be summed without the cross-root action Hessian. The present result computes that cross-root Hessian on the first finite production family for which no source is reused.

Let $\mathcal{P}$ be a finite set of nonzero admissible dual-lattice momentum classes modulo $p \sim -p$. Require the whole collection

$$\{[p], [2p] : [p] \in \mathcal{P}\} \tag{1.1}$$

to be pairwise disjoint and retained by the cutoff. Reveal every source $[p]$ before its target $[2p]$ and set

$$h_{[2p]} = t_p H_p \xi_{[p]}, \qquad H_p \in \mathbb{R}^{12\times 12}, \tag{1.2}$$

with every other control zero. The $\xi_{[p]}$ are independent whitened both-sign roots. Work at $t = 0$ in the exact centered stationary common-even covariance-matched A1/A6/A7 law, with common-even scalar regulator multipliers of modulus at most one.

For the one complete covariance-normal A7 endpoint, the Cameron–Martin source cost, and the terminal sixth power, the origin Hessian satisfies

$$\boxed{D^2\mathcal{A}(0)[H,H] > \frac{1}{40}\sum_{[p]\in\mathcal{P}} \|H_p\|_{\mathrm{HS}}^2, \qquad H \neq 0.} \tag{1.3}$$

This is one global block-row inequality after every endpoint cross block has been retained. It is not an inference from separate two-pair tests.

The theorem is finite-cutoff, positive-floor, fixed-law, affine, and local at the zero control. It excludes nonzero revealed past, overlapping pairs, root reuse, nonlinear or revisit feedback, finite-amplitude convexity, a historical complete-low owner, and cutoff or floor removal. T-050, A13, Nelson, phase selection, every PDE verdict, and Sector A remain open.

2. One scalar endpoint and all covariance responses

For each $[p] \in \mathcal{P}$, let S_p, S_{2p} be the R-151 six-real both-sign synthesis maps and put $D_{p,i} = \partial_i S_p$. At one spatial point the field and current paths are

$$W_t = W_0 + \sum_p t_p S_{2p} H_p \xi_p, \qquad V_{t,i} = V_{0,i} + \sum_p t_p D_{2p,i} H_p \xi_p. \tag{2.1}$$

The disjointness in (1.1) makes all controlled source blocks independent. At the covariance-matched common-even base,

$$Q(0) = Q_{\mathrm{A7}}, \qquad K_i(0) = \mathrm{Cov}(W_0, V_{0,i}) = 0. \tag{2.2}$$

Write $C = \mathrm{Cov}(W)$, $Q = \sum_i \mathrm{Cov}(V_i)$, and $K_i = \mathrm{Cov}(W, V_i)$. The first responses in direction H_p are

$$\begin{aligned} C_p &= S_{2p} H_p S_p^* + S_p H_p^* S_{2p}^*, \\ Q_p &= \sum_i (D_{2p,i} H_p D_{p,i}^* + D_{p,i} H_p^* D_{2p,i}^*), \\ K_{p,i} &= S_{2p} H_p D_{p,i}^* + S_p H_p^* D_{2p,i}^*. \end{aligned} \tag{2.3}$$

There is no mixed second covariance response between distinct p, q because their Gaussian roots are independent. The diagonal second response $Q_{2,p} \succeq 0$ remains in the diagonal block and is never discarded with a negative sign.

Let $B(w) = C_6(w)^* C_6(w)$ and $\overline{T}_{abmn}(C) = \mathbb{E}_C \partial_{ab} B_{mn}(W)$. The exact reduced endpoint is the same scalar as R-151:

$$2\mathcal{R}(t) = \int \left\{ \mathrm{Tr}[(Q(t) - Q_{\mathrm{A7}})\overline{B}(C(t))] + \sum_i (K_i(t))_{am} (K_i(t))_{bn} \overline{T}_{abmn}(C(t)) \right\} dx. \tag{2.4}$$

For $p \neq q$, direct polarization gives the complete mixed block

$$\boxed{\begin{aligned} D_{pq}^2 \mathcal{R}(0) = \int \Big\{ &\frac{1}{4} [(Q_p)_{mn} (C_q)_{ab} + (Q_q)_{mn} (C_p)_{ab}] \overline{T}_{abmn} \\ &+ \frac{1}{2} \sum_i [(K_{p,i})_{am} (K_{q,i})_{bn} + (K_{q,i})_{am} (K_{p,i})_{bn}] \overline{T}_{abmn} \Big\} dx. \end{aligned}} \tag{2.5}$$

The coefficients 1/4 and 1/2 polarize back to the R-151 diagonal coefficients. There is no omitted $C_p C_q$ coefficient-curvature term because $Q(0) - Q_{\mathrm{A7}} = 0$, and no omitted derivative of $\overline{T}$ because $K(0) = 0$. These cancellations are scope hypotheses, not generic identities away from the declared base.

Equation (2.5) is already self-adjoint: $D_{qp}^2 \mathcal{R} = (D_{pq}^2 \mathcal{R})^*$. The legal reverse is this adjoint face, not a second payment. Forest, future variance, balanced, trace, and returned-mean formulae are alternate coordinates or estimates of (2.4); none is appended as an extra reserve.

3. Fourier selection produces a factor-three path graph

Every entry of $C_p, Q_p, K_{p,i}$ has spatial Fourier support

$$\{\pm p, \pm 3p\}. \tag{3.1}$$

The baseline covariance is stationary, so $\overline{T}(C(0))$ is independent of x. Spatial integration in (2.5) therefore vanishes unless a frequency in $\{\pm p, \pm 3p\}$ cancels one in $\{\pm q, \pm 3q\}$. Because (1.1) removes $[q] = [p]$, the only distinct possibilities are

$$[q] = [3p] \quad \text{or} \quad [p] = [3q]. \tag{3.2}$$

Join two vertices of $\mathcal{P}$ exactly when (3.2) holds. Orient each edge from the smaller norm to the larger norm. A vertex has at most one child $[3p]$ and at most one parent $[p/3]$. The norm grows by the strict factor three along every oriented edge, so no cycle exists. Each finite component is consequently a path. This sparse graph is derived from the complete mixed owner; it is not an assumed source orthogonality of the endpoint.

4. Exact off-diagonal owner bound

For $\|H_p\|_{\mathrm{HS}} = 1$ set

$$a_p = \|S_p\|_{\mathrm{op}}, \quad b_p = \|S_{2p}\|_{\mathrm{op}}, \quad d_p = |p| a_p, \quad e_p = 2|p| b_p. \tag{4.1}$$

Then

$$\|C_p\|_{\rm op} \le 2a_pb_p, \quad \|Q_p\|_{\rm op} \le 2d_pe_p, \quad \|K_p\|_{\rm HS} \le b_pd_p + a_pe_p = 3|p|a_pb_p. \tag{4.2}$$

R-130 gives

$$\left\|\frac{1}{2}D^2(B:Q)\right\|_{\rm op} \le H_6\|Q\|_{\rm op}, \qquad H_6 < \frac{7083}{2000}. \tag{4.3}$$

Using the six-real nuclear bound in the Q–C part of (2.5) and the polarized R-151 K–K bound gives

$$|D^2_{pq}\mathcal{R}(0)| \le H_6\{24(|p|^2 + |q|^2) + 108|p||q|\}a_pb_pa_qb_q\,\|H_p\|_{\rm HS}\|H_q\|_{\rm HS}. \tag{4.4}$$

For $[q] = [3p]$, the two braces contribute $240 + 324 = 564$.

Let

$$f(x) = x^2 + Zx + r + \frac{7}{250}, \qquad x = |p|^2. \tag{4.5}$$

The two antipodal signs imply

$$a_p^2 \le \frac{2}{Vf(x)}, \qquad b_p^2 \le \frac{2}{Vf(4x)}. \tag{4.6}$$

After spatial integration, every edge from $[p]$ to $[3p]$ therefore has block norm strictly below

$$E(x) = \frac{2256(7083/2000)x}{V\sqrt{f(x)f(4x)f(9x)f(36x)}}. \tag{4.7}$$

There is no missing factor two in (4.7). The full quadratic form contains $2D^2_{pq}\mathcal{R}[H_p, H_q]$, and $2\|H_p\|\|H_q\| \le \|H_p\|^2 + \|H_q\|^2$ charges the edge weight once to each incident row.

The R-151 diagonal block, including its nonnegative $Q_{2,p}$ term, obeys

$$D^2_{pp}\mathcal{R}(0) > -L(x)\|H_p\|^2_{\rm HS}, \qquad L(x) = \frac{624(7083/2000)x}{Vf(x)f(4x)}. \tag{4.8}$$

5. Rational half-line certificates

Here $V = 4096$. Put

$$A = \frac{624(7083/2000)}{4096}, \qquad D = \frac{2256(7083/2000)}{4096}, \quad F_{14} = f(x)f(4x), \quad F_{936} = f(9x)f(36x). \tag{5.1}$$

The elementary bound $\pi > 31/10$ gives every nonzero dual-lattice mode

$$x = (2\pi/16)^2|n|^2 > \frac{3}{20}. \tag{5.2}$$

If $[p]$ has a factor-three parent, then $p = 3q$ for another nonzero lattice mode and hence $x > 27/20$.

For a vertex without a parent it suffices to prove

$$L(x) + E(x) < \frac{7}{8}. \tag{5.3}$$

The R-151 certificate $L < 4/5$ gives the required sign guard $(7/8)F_{14} - Ax > 0$. Squaring (5.3) is therefore equivalent to positivity of the degree-twelve rational polynomial

$$P_0(x) = \{(7/8)F_{14} - Ax\}^2F_{936} - D^2x^2F_{14}. \tag{5.4}$$

Its exact Sturm chain has length thirteen. At 3/20 and positive infinity the sign variations are both six; the lower-end value is positive. Thus it has no root on $[3/20, \infty)$ and (5.3) holds strictly.

For a vertex with a parent, exact rational certificates give

$$L(x) < \frac{1}{30}, \qquad E(x) < \frac{1}{1000} \quad (x \geq 27/20), \qquad E(y) < \frac{1}{6} \quad (y \geq 3/20). \tag{5.5}$$

The three cleared polynomials have degrees $4, 8, 8$ respectively, are positive at their rational lower endpoints, and have equal Sturm variations at the lower endpoint and positive infinity: $1, 2, 3$ respectively. Hence no one has a root on its declared half-line.

An independent implementation does not reuse a Sturm chain. It maps $x = a + y/(1-y)$, multiplies by $(1-y)^d$, and verifies exact positive Bernstein coefficients on dyadic covers. The degree-twelve cover is

$$[0, 1/16],\ [1/16, 1/8],\ [1/8, 1/4],\ [1/4, 1/2],\ [1/2, 1], \tag{5.6}$$

while the other certificates need at most four intervals. All arithmetic in both implementations is rational.

6. One global block-row estimate

The Cameron–Martin Hessian is diagonal because every source root is used once:

$$D^2\mathcal{A}_{\mathrm{CM}}(0)[H, H] = \frac{9}{10}\sum_p \|H_p\|_{\mathrm{HS}}^2. \tag{6.1}$$

The complete terminal sixth-power Hessian is the single form

$$\frac{9}{10}\int \mathbb{E}\left[|W_0|^4|z_H|^2 + 4|W_0|^2(W_0 \cdot z_H)^2\right] dx \succeq 0, \tag{6.2}$$

where $z_H = \sum_p S_{2p}H_p\xi_p$. We drop (6.2) in its entirety. No diagonal sextic payment is retained while deleting its mixed blocks.

For a graph vertex $[p]$, charge (4.7) once for each incident edge. If it has no parent, (5.3) gives the row margin

$$\frac{9}{10} - L(x) - E(x) > \frac{1}{40}. \tag{6.3}$$

If it has a parent, include both the possible child edge $E(x)$ and the parent edge $E(x/9)$. Equations (5.5) give

$$L(x) + E(x) + E(x/9) < \frac{1}{30} + \frac{1}{1000} + \frac{1}{6} = \frac{201}{1000}, \tag{6.4}$$

so its margin is greater than $699/1000$, in particular greater than $1/40$. Summing the block-row inequalities over every finite path proves (1.3).

This is precisely the global Loewner/Gershgorin step that R-152 required. The proof never replaces it by a collection of positive 2×2 tests.

7. Devil's-advocate review

1. **Objection: pairwise positivity is being summed again. DISMISSED.** Equation (2.5) derives every mixed block from the one scalar endpoint. Equations (3.2) and (6.3)–(6.4) prove one global block-row bound on the resulting matrix.

2. **Objection: the mixed Hessian omits covariance-curvature or K-derivative terms. VALID WITH SCOPE MITIGATION.** Their absence uses exactly $Q(0) = Q_{\mathrm{A7}}$, $K(0) = 0$, independent disjoint roots, and an affine fixed-law chart. Weakening any of those hypotheses requires the additional terms and is outside R-154.

3. **Objection: the 2256 edge coefficient misses the quadratic off-diagonal factor two. DISMISSED.** The factor is $4 \times (240 + 324)$: four is the antipodal covariance normalization. The quadratic factor two is paid by $2ab \leq a^2 + b^2$, once on each endpoint row.
4. **Objection: mixed sixth-power entries need not be positive. DISMISSED.** No individual entry is assigned a sign. The complete global sixth-power Hessian (6.2) is positive semidefinite and is dropped as one form.
5. **Objection: the result supplies the historical complete-low owner or arbitrary predictable feedback. UPHELD AS A BOUNDARY.** The minimal full-output chart declares no independent low coordinate, and it is linear, disjoint, and evaluated only at the origin. No Feshbach block, nonlinear connection, or full T-050 conclusion is claimed.

8. Reproduction, route decision, and Result footer

From the repository root, run

```
E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/
 a13_classii_disjoint_multipair_factor_three_resonance_gap_boundary.py
E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/
 a13_classii_disjoint_multipair_factor_three_resonance_gap_boundary_independent.py
E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/
 a13_classii_disjoint_multipair_factor_three_resonance_gap_boundary_verify.py
```

R-154 is a genuine advance beyond R-151/R-152: a nontrivial finite family of actual cross-root blocks is now evaluated and aggregated without source reuse. The next honest direct step is to relax exactly one load-bearing restriction at a time: first permit one overlapping source/target chain and derive its full covariance second jet; then test nonzero past or nonlinear feedback. A genuine independent low variable enters only if it is declared by that same scalar action. The phase-neutral T-054 route remains separate; R-154 is evidence about a restricted action Hessian, not evidence for BCC or for any alternative phase.

```
Result ID: A13-CLASSII-DISJOINT-MULTIPAIR-FACTOR-THREE-RESONANCE-GAP-
  BOUNDARY (R-154).
Claim: A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION.
Precise statement: On every declared finite disjoint affine p:2p chart, the
  complete mixed endpoint has only factor-three edges and the augmented
  zero-control origin Hessian is strictly above (1/40) sum ||H_p||_F^2.
Scope: Fixed cutoff and floor, nonzero retained dual-lattice modes, pairwise-
  disjoint antipodal source/target classes, centered stationary common-even
  covariance-matched base, affine one-shot controls, and origin Hessian only.
Dependencies: A1, A7, R-130, R-151--R-153, and Sections 1--2 hypotheses.
Evidence grade: ANALYTIC, EXACT, EXECUTED, AUDITED.
Reproduction command: E:\Dev\TECT.venv\Scripts\python.exe
  codes/foundations/a13_classii_disjoint_multipair_factor_three_resonance_
  gap_boundary_verify.py
Expected output: Integrated 157/157 PASS including primary 34/34,
  independent 26/26, all 60 embedded child rows, 97 integrator-only checks,
  and every artifact/PDF/generated-surface gate.
Falsification gate: Failure of the mixed coefficients, Fourier support,
  factor 2256, any Sturm or Bernstein half-line certificate, source/sextic
  ownership, authority hash, executable/JSON/PDF, exploration, or surface pin.
Tier before / after: T4 / T4; no promotion.
No-overclaim statement: No nonzero-past, overlapping/revisit, nonlinear or
  full progressive theorem; no T-050/A13, Nelson/measure, phase/PDE verdicts, or
  Sector-A conclusion follows.
Next required action: Derive one overlapping source/target affine chain from
  the same scalar action before testing nonzero past or nonlinear feedback;
  add low/Feshbach only when a genuine independent low coordinate exists.
```